

## The Ergonomic Web

There are plenty of online resources that will give you all the information you need about ergonomics, and more. Most of these Web sites are dedicated to the science of ergonomics, and you can be sure that the information provided is correct. Here's a list of a few sites you should visit:

### The Ergonomics Society ([www.ergonomics.org.uk](http://www.ergonomics.org.uk))

This is a society of people dedicated to the study of ergonomics. You can find everything from information to job vacancies for ergonomic specialists here.

### Office Ergonomics ([www.office-ergo.com](http://www.office-ergo.com))

This is a site that's dedicated to providing you information on how you can be more ergonomic at work, from tips for employees to guidelines for employers.

### Ergonomics: Posture Movement and Ergonomics ([www.ergonomics.org](http://www.ergonomics.org))

This site covers all the do's and don't's of office behaviour, including the correct postures and monitor placement.

### Cornell University Ergonomics Web (<http://ergo.human.cornell.edu/>)

This site has information from research studies and class work by students and faculty from the Cornell Human Factors and Ergonomics Research Group.

### International Ergonomics Association ([www.iea.cc](http://www.iea.cc))

Perhaps the largest ergonomics society, the IEA site will give you details on conferences and technical contacts in case you want to get your products certified as ergonomic-friendly.

the problem of resolutions. Placing a 19-inch monitor three feet away from the eyes might seem like a good idea, but when you're running at a resolution of 1600 x 1280, this could make the text on the screen look minuscule, and force you to strain your eyes anyway!

A good practice is to keep the monitor at least 25 inches away from the eyes, and adjust to the resolution that causes the least eye strain. Instead of using a 19-inch monitor running at a resolution of 1600 x 1280, placed at 25 inches away from the eyes, consider placing the monitor three feet away, and running it at a resolution of 1280 x 1024 or 1024 x 768.

The next consideration is the monitor tilt, and the height at which it is placed. Here's where your table comes into play. Since you've already decided on the height of your chair, you need to make sure the tables are low enough to keep the monitor at the correct height. Remember, it's a lot easier to increase the height of a monitor by placing something under it than it is to decrease the height.

Since heights are very subjective, monitor placement is always decided by angles. Studies have found that in order to protect the neck, a monitor's viewing area should be 15 to 50 degrees below horizontal eye level. This means that if you sit in your chair and look straight ahead, focusing on an object at the same height above the ground as your eyes, the top of your monitor should be at a minimum of 15 degrees below your line of sight, while the bottom of the monitor screen should be a maximum of 50 degrees below the line of sight. This is illustrated in the diagram alongside (see *Monitor Placement*).

A few other things to consider are the results of studies that have shown that white text on black backgrounds seem to strain the eyes more than black text on white backgrounds. So as a

rule of thumb, try and work with dark text on light backgrounds. Keep in mind that correct ergonomic placement of the monitor requires the monitor to be tilted away from the eyes, so that the bottom of the screen is closer to the body than the top of the screen.

To better understand why a tilt is needed, notice the way you're holding this copy of *Digit*. You'll see that the magazine is tilted away from you, just as when you hold anything else, say, a newspaper or book. This happens because we're used to objects being further away when at the eye level and closer when below our line of sight.

If you sit down and look straight ahead, you will notice that the objects at and above your line of sight are further away, and the objects in your field of vision that are below this line of sight are much closer. It's only obvious that if you tilt your monitor away from yourself to reflect a more real-world sight, you will strain your eyes less.

## Keyboard And Mouse Placement

The next step towards protecting your body when working with computers is to protect your hands—your fingers, wrist, forearms and elbows. As we mentioned earlier, the most common injury resulting from computer use is the Carpal Tunnel Syndrome, caused by excessive use of a keyboard. Though a keyboard that solves this problem has not yet been invented, especially for those who have to type hundreds of thousands of words a day (data-entry operators and the like), there are models that try and reduce the risks.

**Split keyboards:** These are keyboards that are broken into two parts. Some designs feature a single keyboard with the keypad split in two—and each half is positioned at a different angle.



Split keyboards attempt to let you place each hand as comfortably as you want to, and at whatever angle you desire

This style of keyboard tries to solve the problem of unnatural angles for the wrists.

To illustrate the problem, look at your regular keyboard. It is flat and has the same angle towards your body for all keys. Now place your hands over the keyboard as you would when beginning to type. You'll notice that your wrists are bent slightly outwards (towards your little fingers). This posture is not natural, and we often find ourselves rotating our keyboards in one direction or the other to ease the stress on our